

Tarkan Zarrouk

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EDUCATION

Carleton University

Bachelor of Mathematics

Ottawa, Ontario

Sep. 2025 – Present

- Relevant coursework: Data Structures and Algorithms, Discrete Mathematics, Calculus, and Computer Systems.
- Recipient of the Presidential Entrance Scholarship valued at \$6,000.

High School

Ontario Secondary School Diploma

Ontario, Canada

Sep. 2021 – Jun. 2025

- Graduated with an 88.75% cumulative average.
- Achieved 98% in Grade 12 Computer Science and 96% in Computer Engineering Technology.
- Recognized for the highest overall average in Computer Engineering Technology.

EXPERIENCE

Frontend Web Developer

Kaie

Feb. 2025 – Apr. 2025

Mississauga, Ontario

- Developed and improved user-facing web application features, including user profile management and an Explore page, to improve personalization and content discoverability.
- Implemented frontend components and integrated application functionality with backend services and data structures.
- Contributed to the migration of 2,000+ user accounts to a new database while maintaining data integrity and minimizing service disruption.

Code Sensei

Code Ninjas

Jul. 2024 – Aug. 2024

Ontario, Canada

- Taught programming concepts to children aged 6–13 through hands-on coding activities and classroom instruction
- Guided students through object-oriented programming, algorithms, debugging, and problem-solving.
- Collaborated with colleagues to deliver age-appropriate lessons and maintain an engaging learning environment.

PROJECTS

Social Media Application | *React, TypeScript, Firebase, JavaScript*

Feb. 2025 – Apr. 2025

- Built a database-driven social media application to develop practical experience with CRUD operations, application architecture, and backend integration.
- Integrated Firebase services to manage application data and support user-facing functionality.
- Implemented data handling and validation to improve reliability when creating, retrieving, updating, and deleting application records.

Computer Vision Detection System | *Python, OpenCV, YOLOv5*

- Developed computer vision prototypes for face and QR code detection using YOLOv5 and OpenCV.
- Worked with trained model weights and image processing pipelines to detect and classify visual features.
- Explored object detection and computer vision techniques for real-world recognition tasks.

Real-Time Tracking System | *Python, MediaPipe, Arduino*

- Experimented with MediaPipe-based tracking and integrated visual input with Arduino-controlled hardware.
- Connected software processing with physical actuator control to explore interactive tracking systems.

Autonomous Wall-Following Robot | *C++, Arduino, Ultrasonic Sensors*

- Programmed an Arduino-powered robot to navigate autonomously using ultrasonic distance sensors and motor-control logic.
- Implemented obstacle detection and wall-following behaviour using sensor feedback and conditional logic.
- Integrated motor drivers and sensors to coordinate movement and respond to environmental conditions.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS, SQL
Frameworks and Technologies: React, Node.js, Firebase, OpenCV, YOLOv5, MediaPipe, Arduino
Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, Linux
Areas of Interest: Software Engineering, Full-Stack Development, Machine Learning, Computer Vision

AWARDS AND CERTIFICATIONS

Presidential Entrance Scholarship <i>Carleton University</i>	2025
Ontario Scholar <i>Ontario</i>	Jun. 2025
Top Mark in Computer Engineering Technology <i>High School Recognition</i>	Jun. 2025
Python (Basic) and React (Basic) <i>HackerRank</i>	Dec. 2025